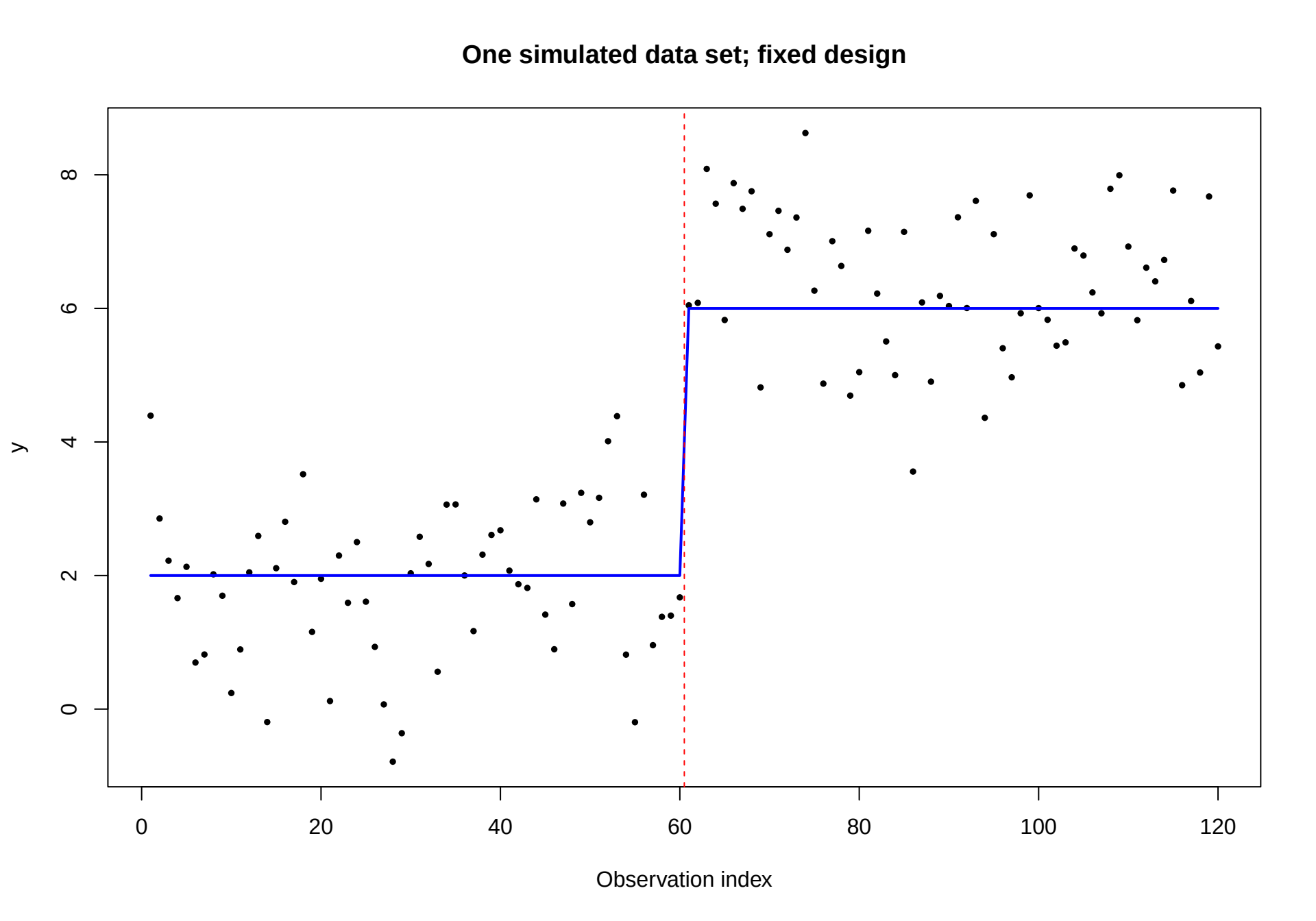
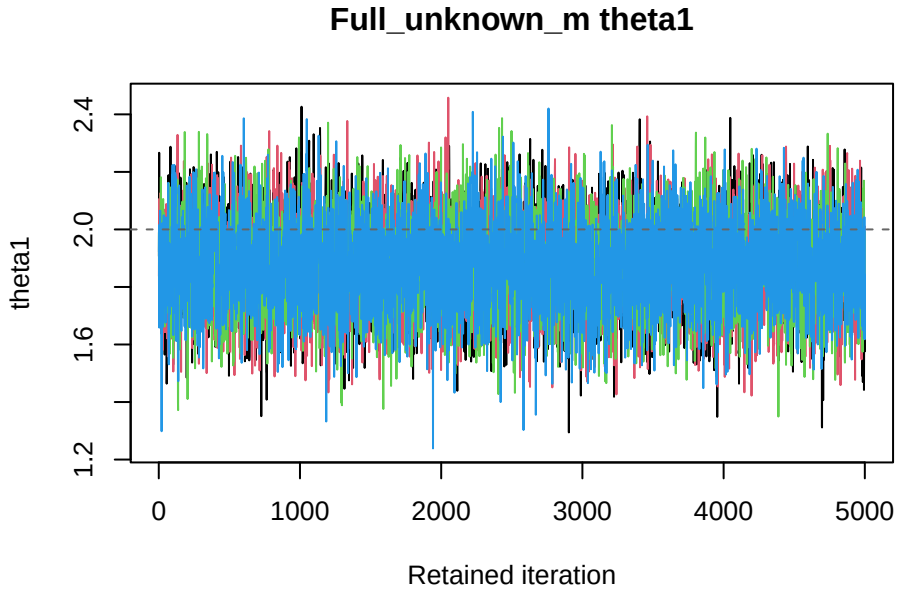
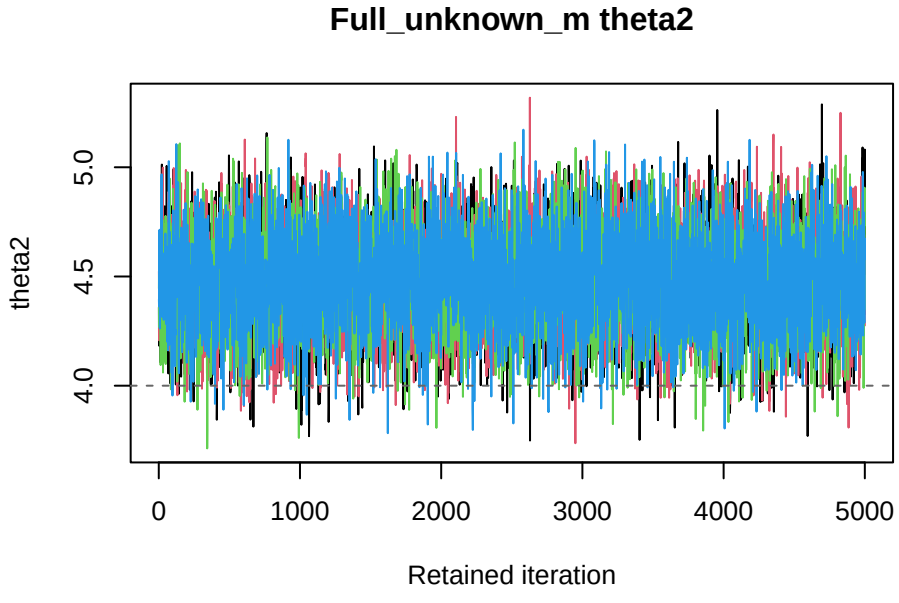
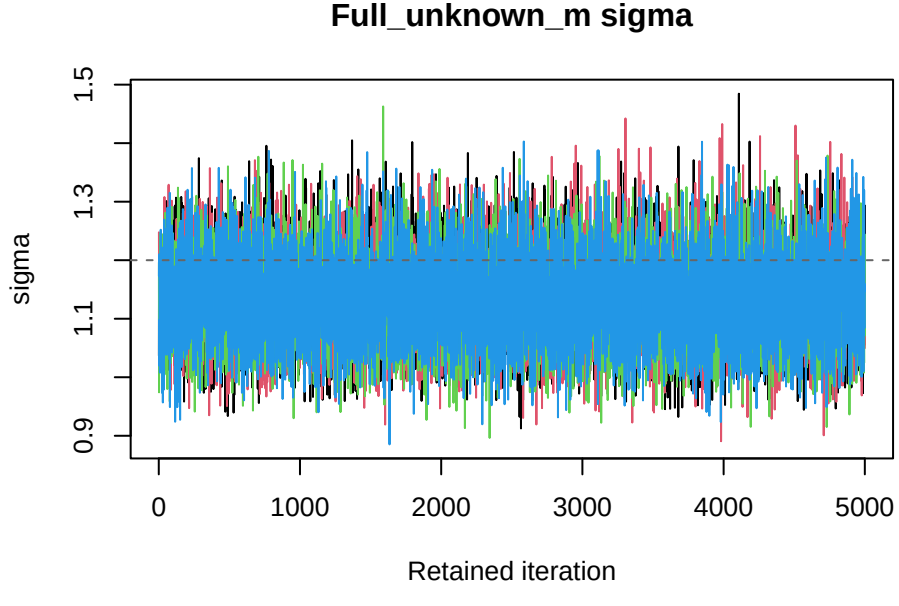
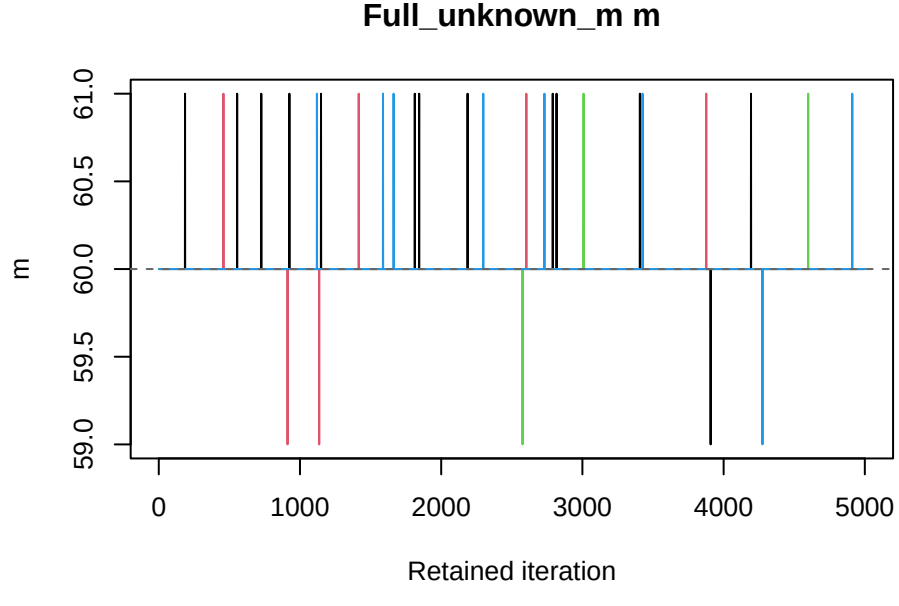
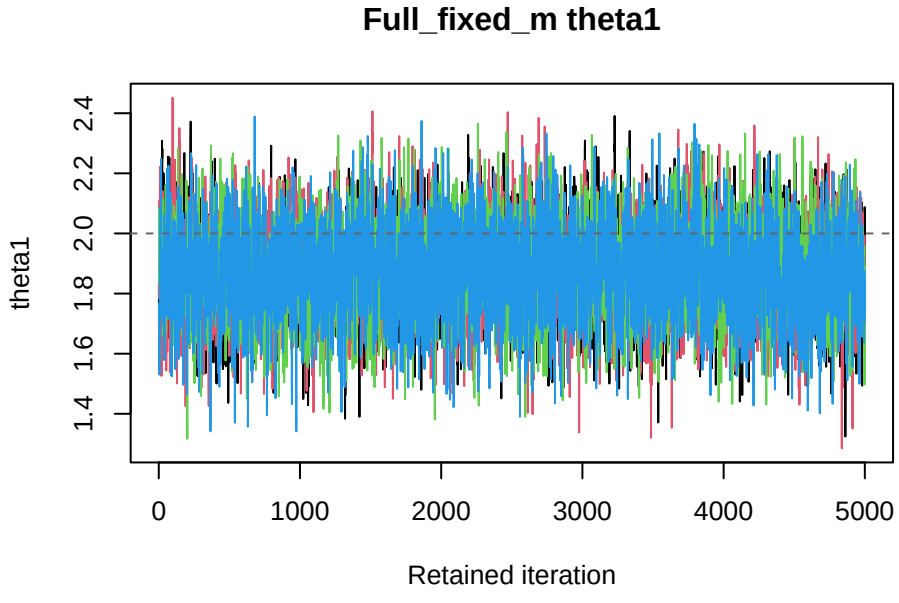

One simulated data set; fixed design
y
0
2
4
6
8
0
20
40
60
80
100
120
Observation index

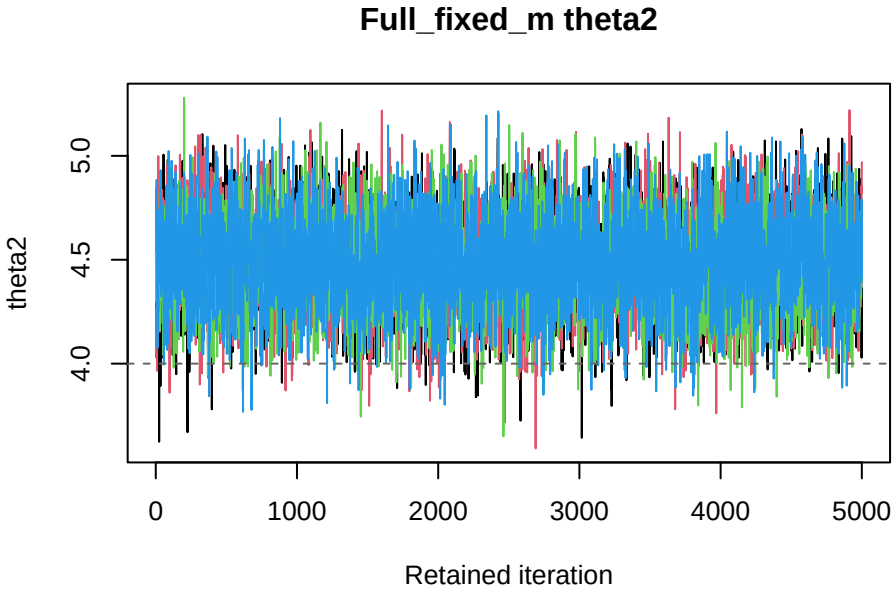

Full_unknown_m theta1
theta1
2.4
2.0
1.6
1.2
0
1000
2000
3000
4000
5000
Retained iteration

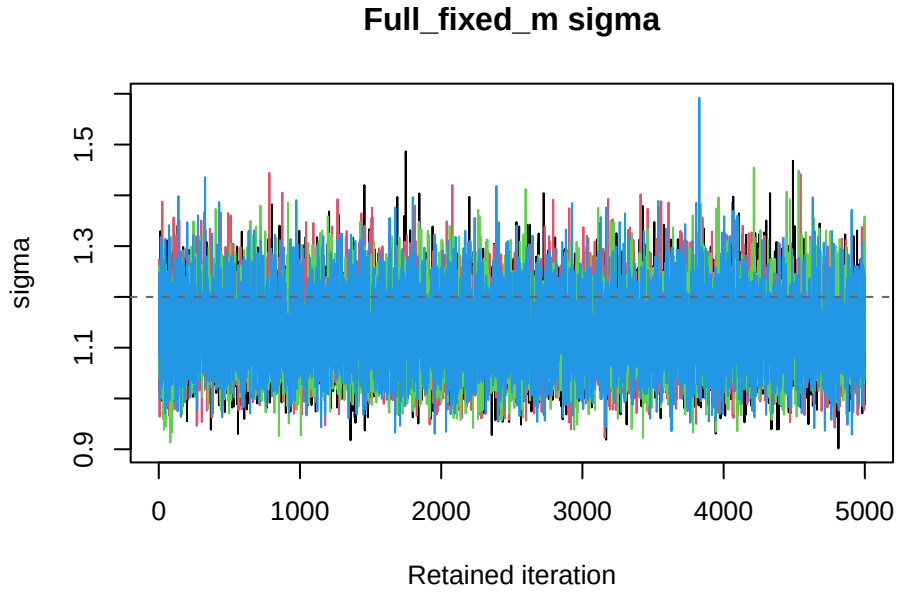

Full_unknown_m theta2
theta2
5.0
4.5
4.0
0
1000
2000
3000
4000
5000
Retained iteration

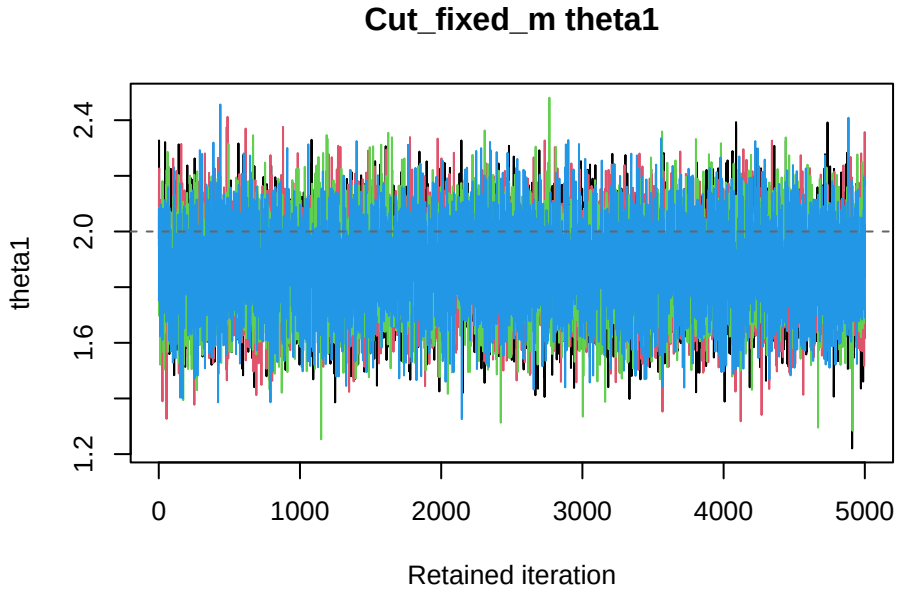

Full_unknown_m sigma
sigma
1.5
1.3
1.1
0.9
0
1000
2000
3000
4000
5000
Retained iteration


Full_unknown_m m
m
61.0
60.5
60.0
59.5
59.0
0
1000
2000
3000
4000
5000
Retained iteration

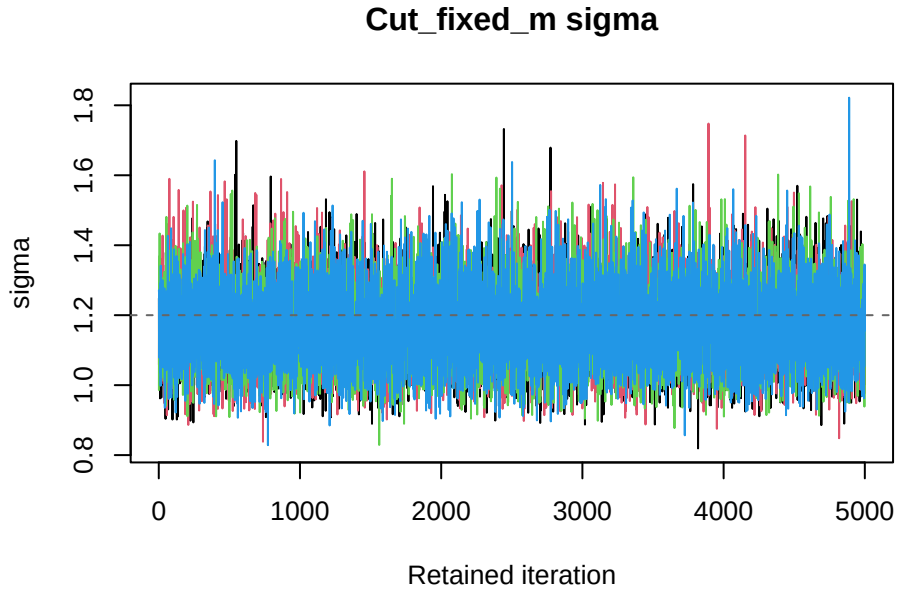

Full_fixed_m theta1
theta1
1.4
1.6
1.8
2.0
2.2
2.4
0
1000
2000
3000
4000
5000
Retained iteration

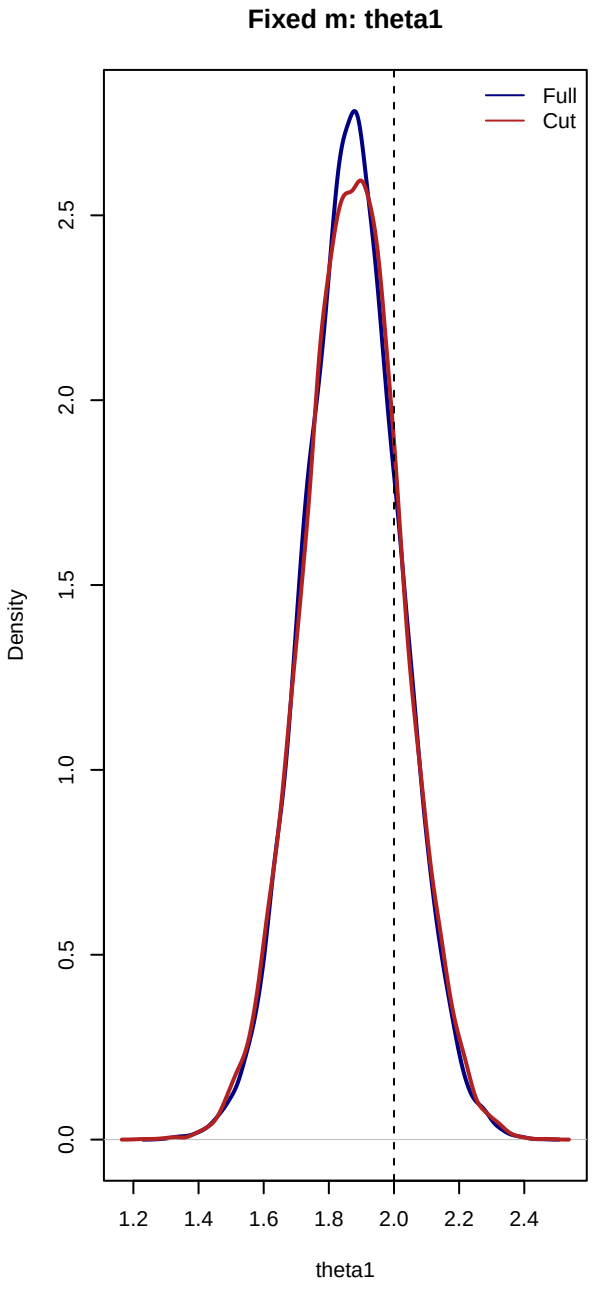

Full_fixed_m theta2
theta2
4.0
4.5
5.0
0
1000
2000
3000
4000
5000
Retained iteration

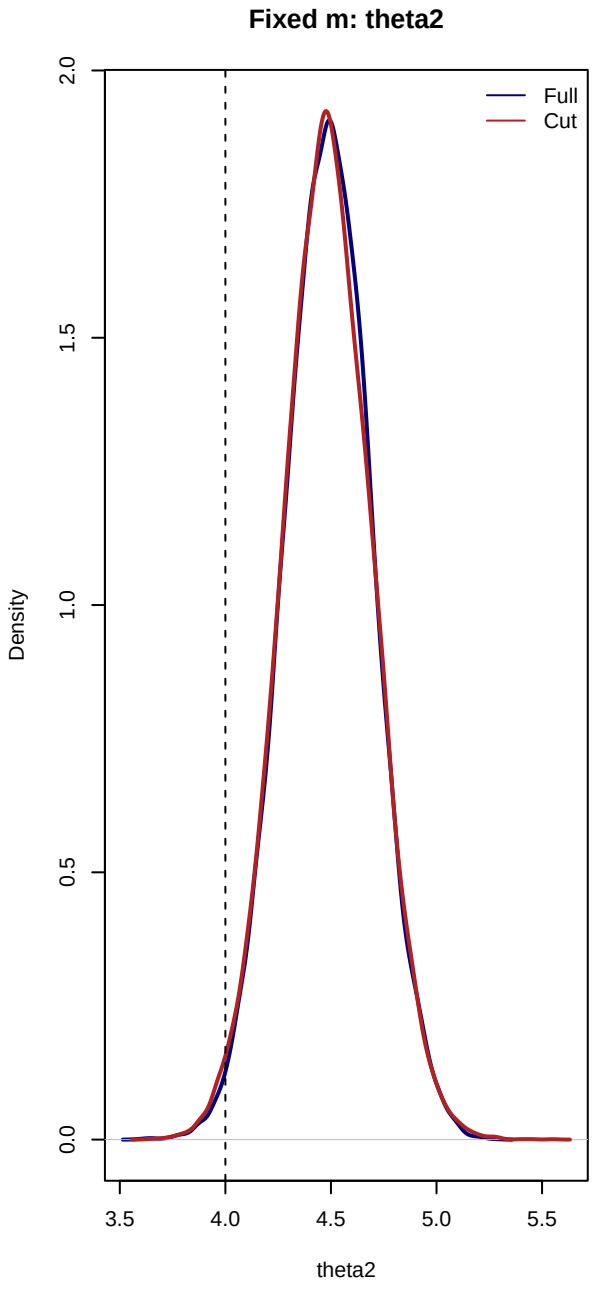

Full_fixed_m sigma
sigma
0.9
1.1
1.3
1.5
0
1000
2000
3000
4000
5000
Retained iteration

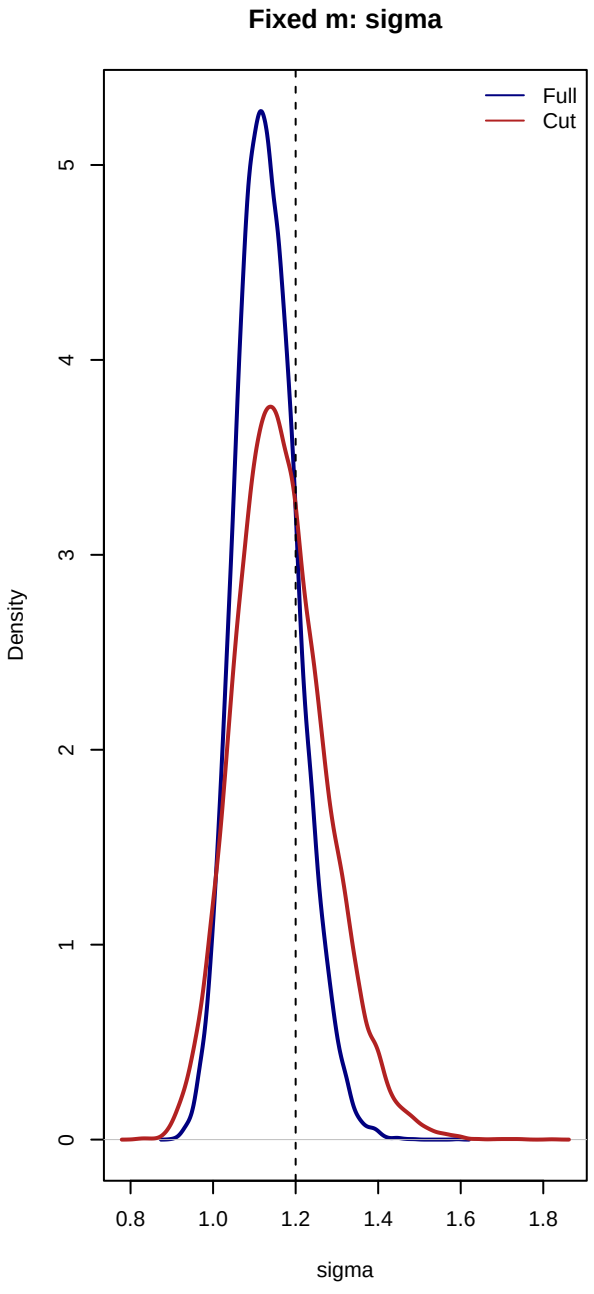

Cut_fixed_m theta1
theta1
2.4
2.0
1.6
1.2
0
1000
2000
3000
4000
5000
Retained iteration

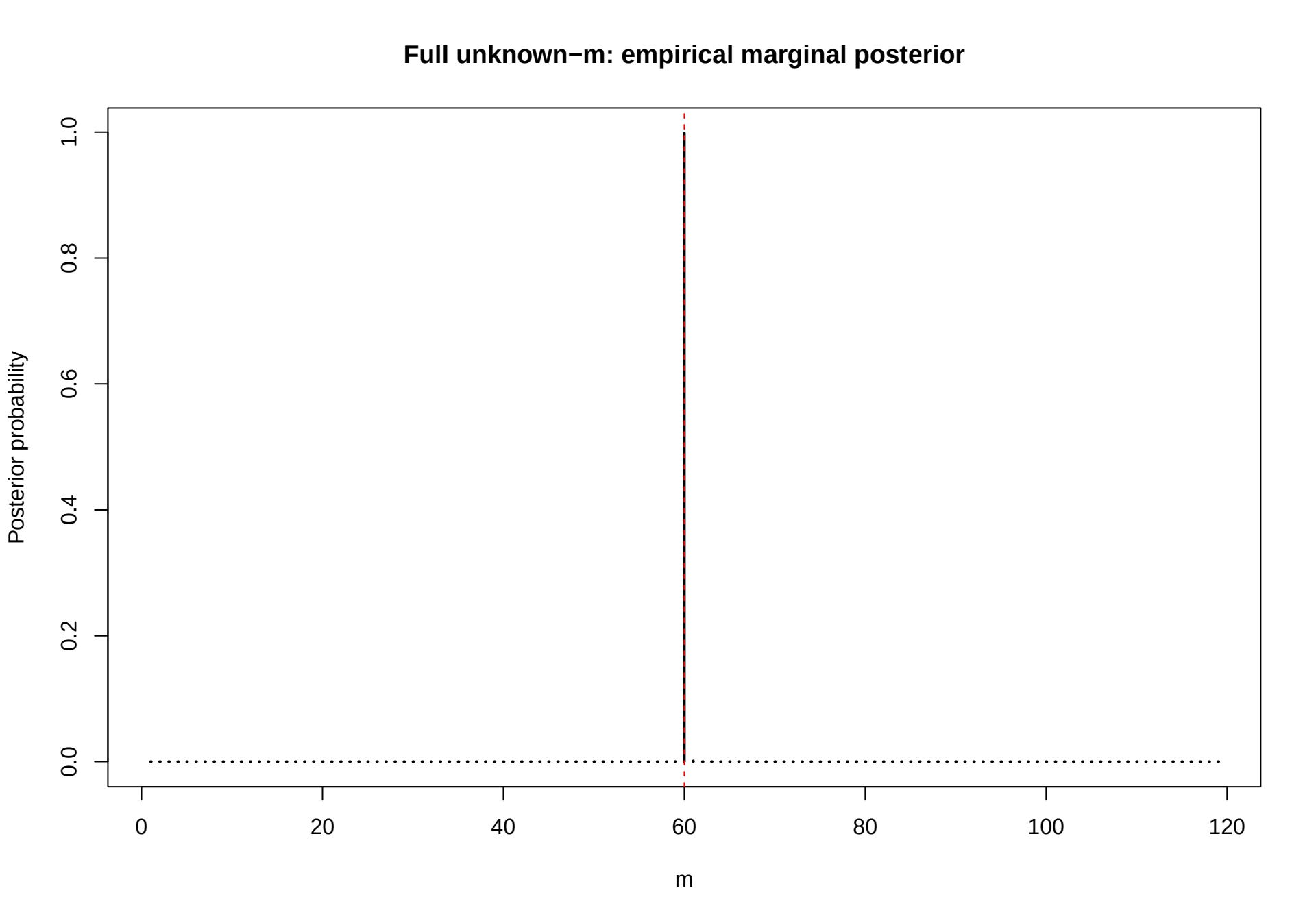

Cut_fixed_m theta2
theta2
5.5
5.0
4.5
4.0
0
1000
2000
3000
4000
5000
Retained iteration


Cut_fixed_m sigma
sigma
1.8
1.6
1.4
1.2
1.0
0.8
0
1000
2000
3000
4000
5000
Retained iteration


Fixed m: theta1
Full
Cut
Density
0.0
0.5
1.0
1.5
2.0
2.5
1.2
1.4
1.6
1.8
2.0
2.2
2.4
theta1


Fixed m: theta2
Full
Cut
Density
0.0
0.5
1.0
1.5
2.0
3.5
4.0
4.5
5.0
5.5
theta2


Fixed m: sigma
Full
Cut
Density
0
1
2
3
4
5
0.8
1.0
1.2
1.4
1.6
1.8
sigma


Full unknown−m: empirical marginal posterior
Posterior probability
0.0
0.2
0.4
0.6
0.8
1.0
0
20
40
60
80
100
120
m